


```

!pip install --quiet jax jaxlib

import jax
import jax.numpy as jnp
import numpy as np
from jax.scipy.linalg import expm

# =====
# SU(3) GENERATORS (anti-Hermitian basis  $T_a = i \lambda_a / 2$ )
# =====

def su3_generators():
    lam = []

    lam1 = jnp.array([[0,1,0],[1,0,0],[0,0,0]], dtype=jnp.complex64)
    lam2 = jnp.array([[0,-1j,0],[1j,0,0],[0,0,0]], dtype=jnp.complex64)
    lam3 = jnp.array([[1,0,0],[0,-1,0],[0,0,0]], dtype=jnp.complex64)
    lam4 = jnp.array([[0,0,1],[0,0,0],[1,0,0]], dtype=jnp.complex64)
    lam5 = jnp.array([[0,0,-1j],[0,0,0],[1j,0,0]], dtype=jnp.complex64)
    lam6 = jnp.array([[0,0,0],[0,0,1],[0,1,0]], dtype=jnp.complex64)
    lam7 = jnp.array([[0,0,0],[0,0,-1j],[0,1j,0]], dtype=jnp.complex64)
    lam8 = jnp.array([[1,0,0],[0,1,0],[0,0,-2]], dtype=jnp.complex64)/jnp.sq

    lam = jnp.stack([lam1,lam2,lam3,lam4,lam5,lam6,lam7,lam8], axis=0)
    T = 1j * lam / 2.0 # anti-Hermitian
    return T

T_SU3 = su3_generators()

def su3_alg_from_vec(a_vec):
    return jnp.einsum("...a,aij->...ij", a_vec, T_SU3)

def su3_exp(a_vec):
    A = su3_alg_from_vec(a_vec)
    return expm(A)

# =====
# LATTICE HELPERS
# =====

def shift_idx(idx, mu, L):
    x,y,z,t = idx
    if mu==0: return ((x+1)%L, y, z, t)
    if mu==1: return (x, (y+1)%L, z, t)
    if mu==2: return (x, y, (z+1)%L, t)
    if mu==3: return (x, y, z, (t+1)%L)
    raise ValueError("mu must be 0,1,2,3")

def all_sites(L):

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        return [(x,y,z,t)
                for x in range(L)
                for y in range(L)
                for z in range(L)
                for t in range(L)]

# =====
# BUILD LINK MATRICES  U = exp(A)
# =====

def build_links_from_params(params, L):
    flat = params.reshape(-1, 8)
    U_flat = jax.vmap(su3_exp)(flat)
    return U_flat.reshape(L,L,L,L,4,3,3)

# =====
# WILSON ACTION (SU(3))
# =====

def wilson_action(params, L, beta):
    U = build_links_from_params(params, L)
    action = 0.0

    for x,y,z,t in all_sites(L):
        for mu in range(4):
            for nu in range(mu+1,4):
                idx = (x,y,z,t)
                x_mu = shift_idx(idx, mu, L)
                x_nu = shift_idx(idx, nu, L)

                U1 = U[x,y,z,t,mu]
                U2 = U[x_mu[0], x_mu[1], x_mu[2], x_mu[3], nu]
                U3 = jnp.conjugate(U[x_nu[0], x_nu[1], x_nu[2], x_nu[3], mu])
                U4 = jnp.conjugate(U[x,y,z,t,nu].T)

                Uplaq = U1 @ U2 @ U3 @ U4
                action += beta * (1.0 - (1.0/3.0)*jnp.real(jnp.trace(Uplaq)))

    return action

# =====
# HAAR MASS TERM (quadratic from -log J(A) ≈ c0 Tr(A†A))
# =====

def haar_mass_term_exact(params, c0):
    flat = params.reshape(-1,8)

    def per_link(a_vec):
        A = su3_alg_from_vec(a_vec)
        return jnp.real(jnp.trace(A.conj().T @ A))

```

```

    tr_vals = jax.vmap(per_link)(flat)
    return c0 * jnp.sum(tr_vals)

def total_action(params, L, beta, c0):
    return wilson_action(params, L, beta) + haar_mass_term_exact(params, c0)

# =====
# FLATTENING FOR HESSIAN
# =====

def make_flat_funcs(L, beta, c0):
    sample = jnp.zeros((L,L,L,L,4,8), dtype=jnp.float32)
    n_params = sample.size

    def unflatten(theta):
        return theta.reshape((L,L,L,L,4,8))

    def flat_action(theta):
        return total_action(unflatten(theta), L, beta, c0)

    return flat_action, unflatten, n_params

# =====
# FULL HESSIAN COMPUTATION
# =====

def compute_hessian(L=2, beta=2.0, c0=0.25):
    flat_action, unflatten, n_params = make_flat_funcs(L, beta, c0)

    print(f"L={L}, n_params={n_params}")

    theta0 = jnp.zeros((n_params,), dtype=jnp.float32)
    H = jax.hessian(flat_action)(theta0)
    return H

# =====
# EXECUTE
# =====

L = 2      # try 2 first, then maybe 3 if GPU is strong
beta = 2.0
c0 = 0.25  # SU(3) Haar coefficient ~ N/12 = 3/12 = 0.25 in chosen basis

print("Computing Hessian for SU(3) Wilson + Haar ...")
H = compute_hessian(L, beta, c0)

print("Hessian shape:", H.shape)

H_np = np.array(H, dtype=np.float64)
evals = np.linalg.eigvalsh(H_np)

```

```
print("Smallest 10 eigenvalues:")
print(evals[:10])
```

```
Computing Hessian for SU(3) Wilson + Haar ...
L=2, n_params=512
/usr/local/lib/python3.12/dist-packages/jax/_src/lax/lax.py:5473: ComplexWarning
  x_bar = _convert_element_type(x_bar, x.aval.dtype, x.aval.weak_type)
Hessian shape: (512, 512)
Smallest 10 eigenvalues:
[0.24999976 0.24999976 0.24999976 0.24999976 0.24999976 0.24999976
 0.24999976 0.24999976 0.24999976 0.24999976]
```

```
# --- keep all previous definitions above this line (su3_generators, su3_alg
#     su3_exp, shift_idx, all_sites, build_links_from_params, wilson_action,
#     haar_mass_term_exact, total_action, make_flat_funcs, compute_hessian)
```

```
import numpy as np
```

```
def compute_hessian_split(L=2, beta=2.0, c0=0.25):
    # 1) Wilson-only Hessian (c0=0)
    flat_action_W, unflatten_W, n_params = make_flat_funcs(L, beta, 0.0)
    theta0 = jnp.zeros((n_params,), dtype=jnp.float32)
    H_W = jax.hessian(flat_action_W)(theta0)
    H_W_np = np.array(H_W, dtype=np.float64)
    evals_W = np.linalg.eigvalsh(H_W_np)
```

```
    # 2) Haar-only Hessian (beta=0)
    flat_action_H, unflatten_H, _ = make_flat_funcs(L, 0.0, c0)
    H_H = jax.hessian(flat_action_H)(theta0)
    H_H_np = np.array(H_H, dtype=np.float64)
    evals_H = np.linalg.eigvalsh(H_H_np)
```

```
    # 3) Wilson+Haar (for sanity, should match previous run)
    flat_action_WH, unflatten_WH, _ = make_flat_funcs(L, beta, c0)
    H_WH = jax.hessian(flat_action_WH)(theta0)
    H_WH_np = np.array(H_WH, dtype=np.float64)
    evals_WH = np.linalg.eigvalsh(H_WH_np)
```

```
print(f"L={L}, n_params={n_params}")
print("Wilson-only Hessian:")
print("  min eigenvalue:", evals_W[0])
print("  max eigenvalue:", evals_W[-1])
```

```
print("\nHaar-only Hessian:")
print("  min eigenvalue:", evals_H[0])
print("  max eigenvalue:", evals_H[-1])
```

```
print("\nWilson + Haar Hessian:")
print("  min eigenvalue:", evals_WH[0])
```

```

    print("  max eigenvalue:", evals_WH[-1])

    return (evals_W, evals_H, evals_WH)

# ---- run split diagnostic ----
L = 2
beta = 2.0
c0 = 0.25

print("Computing split Hessians (Wilson-only, Haar-only, Wilson+Haar)...")
evals_W, evals_H, evals_WH = compute_hessian_split(L=L, beta=beta, c0=c0)

```

Computing split Hessians (Wilson-only, Haar-only, Wilson+Haar)...

L=2, n_params=512

Wilson-only Hessian:

min eigenvalue: -2.38418584430633e-07

max eigenvalue: 5.333333730697644

Haar-only Hessian:

min eigenvalue: 0.2499999850988388

max eigenvalue: 0.25

Wilson + Haar Hessian:

min eigenvalue: 0.24999976158141646

max eigenvalue: 5.5833337306976425

```

import jax
import jax.numpy as jnp
import numpy as np

# -----
# 1. Helper: sample random config in A-space
# -----

def sample_random_theta(L, scale, key):
    """
    Sample a random configuration in parameter space (A-coordinates).
    params shape: (L,L,L,L,4,8), flattened to theta.
    Each component ~ Normal(0, scale).
    """
    sample = scale * jax.random.normal(key, (L, L, L, L, 4, 8), dtype=jnp.float64)
    return sample.reshape(-1)

# -----
# 2. Compute lambda_min at a given config
# -----

def lambda_min_at_theta(theta_flat, L, beta, c0):
    """
    Compute smallest eigenvalue of Hessian of:

```

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        - Wilson-only (c0=0)
        - Wilson+Haar (given c0)
    at the given configuration theta_flat.
    Returns (lambda_min_Wilson, lambda_min_WilsonHaar).
    """
    # Wilson-only
    flat_action_W, unflatten_W, n_params = make_flat_funcs(L, beta, 0.0)
    assert theta_flat.shape[0] == n_params
    H_W = jax.hessian(flat_action_W)(theta_flat)
    H_W_np = np.array(H_W, dtype=np.float64)
    evals_W = np.linalg.eigvalsh(H_W_np)
    lam_min_W = float(evals_W[0])

    # Wilson + Haar
    flat_action_WH, unflatten_WH, _ = make_flat_funcs(L, beta, c0)
    H_WH = jax.hessian(flat_action_WH)(theta_flat)
    H_WH_np = np.array(H_WH, dtype=np.float64)
    evals_WH = np.linalg.eigvalsh(H_WH_np)
    lam_min_WH = float(evals_WH[0])

    return lam_min_W, lam_min_WH

# -----
# 3.  $\beta$ -scan over random configs
# -----

def scan_betas_random_configs(L=2,
                              betas=(0.5, 1.0, 2.0),
                              c0=0.25,
                              n_samples=3,
                              scale=0.1,
                              seed=1234):
    """
    For each beta in `betas`, draw `n_samples` random configs with amplitude
    compute lambda_min(Wilson) and lambda_min(Wilson+Haar), and print a tabl
    """
    key = jax.random.PRNGKey(seed)
    flat_action_tmp, unflatten_tmp, n_params = make_flat_funcs(L, betas[0],

    print(f"L={L}, n_params={n_params}")
    print(f"Random config scale = {scale}")
    print(f"c0 (Haar coeff) = {c0}")
    print(f"Number of samples per beta = {n_samples}\n")

    for beta in betas:
        print(f"=== beta = {beta} ===")
        for s in range(n_samples):
            key, subkey = jax.random.split(key)
            theta_flat = sample_random_theta(L, scale, subkey)

            lam W. lam WH = lamhda min at theta(theta_flat, L, beta, c0)

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print(f" sample {s}: lambda_min(W) = {lam_W:+.6f} , "
      f"lambda_min(W+Haar) = {lam_WH:+.6f}")
print("")

# -----
# 4. Run the scan
# -----

L = 2
betas = (0.5, 1.0, 2.0) # adjust as desired
c0 = 0.25                # Haar coefficient
n_samples = 3            # per beta
scale = 0.1              # random A amplitude

scan_betas_random_configs(L=L,
                           betas=betas,
                           c0=c0,
                           n_samples=n_samples,
                           scale=scale,
                           seed=2025)

```

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L=2, n_params=512
Random config scale = 0.1
c0 (Haar coeff) = 0.25
Number of samples per beta = 3

=== beta = 0.5 ===
sample 0: lambda_min(W) = -0.058897 , lambda_min(W+Haar) = +0.191103
sample 1: lambda_min(W) = -0.065420 , lambda_min(W+Haar) = +0.184580
sample 2: lambda_min(W) = -0.082270 , lambda_min(W+Haar) = +0.167730

=== beta = 1.0 ===
sample 0: lambda_min(W) = -0.137351 , lambda_min(W+Haar) = +0.112649
sample 1: lambda_min(W) = -0.133411 , lambda_min(W+Haar) = +0.116589
sample 2: lambda_min(W) = -0.143963 , lambda_min(W+Haar) = +0.106037

=== beta = 2.0 ===
sample 0: lambda_min(W) = -0.270086 , lambda_min(W+Haar) = -0.020086
sample 1: lambda_min(W) = -0.277344 , lambda_min(W+Haar) = -0.027344
sample 2: lambda_min(W) = -0.311023 , lambda_min(W+Haar) = -0.061023

```

```

import jax
import jax.numpy as jnp
import numpy as np

def estimate_beta_critical(L=2,
                           betas=jnp.linspace(0.2, 2.2, 11),
                           c0=0.25,
                           n_samples=5

```

```

        n_samples=5,
        scale=0.1,
        seed=999):
    """
    For each beta, draw n_samples random configs of amplitude `scale`,
    compute min eigenvalue of Hessian for Wilson+Haar, take the *worst* (min
    Return a table (beta, min_over_samples lambda_min(W+Haar)).
    """
    key = jax.random.PRNGKey(seed)
    flat_action_tmp, unflatten_tmp, n_params = make_flat_funcs(L, float(beta)

    results = []
    for beta in np.array(betas):
        beta = float(beta)
        lam_mins = []
        for s in range(n_samples):
            key, subkey = jax.random.split(key)
            theta_flat = sample_random_theta(L, scale, subkey)
            # Here we only care about W+Haar
            flat_action_WH, unflatten_WH, _ = make_flat_funcs(L, beta, c0)
            H_WH = jax.hessian(flat_action_WH)(theta_flat)
            H_WH_np = np.array(H_WH, dtype=np.float64)
            evals_WH = np.linalg.eigvalsh(H_WH_np)
            lam_mins.append(float(evals_WH[0]))
        lam_mins = np.array(lam_mins)
        worst = lam_mins.min()
        results.append((beta, worst))

    print(f"L={L}, scale={scale}, c0={c0}, n_samples={n_samples}")
    print("beta      min_lambda_min(W+Haar) over samples")
    for beta, worst in results:
        print(f"{beta:4.1f}      {worst:+.6f}")

    return results

# ---- run critical beta scan ----
L = 2
betas = jnp.linspace(0.2, 2.2, 11) # 0.2,0.4,...,2.2
c0 = 0.25
n_samples = 5
scale = 0.1

beta_scan = estimate_beta_critical(L=L,
                                   betas=betas,
                                   c0=c0,
                                   n_samples=n_samples,
                                   scale=scale,
                                   seed=2026)

```

```
L=2, scale=0.1, c0=0.25, n_samples=5
```

beta	min_lambda_min(W+Haar) over samples
0.2	+0.221226
0.4	+0.191768
0.6	+0.169219
0.8	+0.137221
1.0	+0.101844
1.2	+0.073269
1.4	+0.049121
1.6	+0.011211
1.8	-0.014185
2.0	-0.063798
2.2	-0.053980

```

import jax
import jax.numpy as jnp
import numpy as np

def estimate_convexity_grid(L=2,
                            betas=jnp.linspace(0.4, 2.0, 9),
                            scales=(0.05, 0.1, 0.15, 0.2),
                            c0=0.25,
                            n_samples=4,
                            seed=4242):
    """
    For each (beta, scale), sample n_samples random configs with amplitude `
    compute lambda_min(W+Haar), and take the worst (most negative).
    Returns a list of (beta, scale, worst_lambda).
    """
    key = jax.random.PRNGKey(seed)
    results = []

    for beta in np.array(betas):
        beta = float(beta)
        for scale in scales:
            lam_mins = []
            for s in range(n_samples):
                key, subkey = jax.random.split(key)
                theta_flat = sample_random_theta(L, scale, subkey)

                # Hessian for W+Haar at this theta
                flat_action_WH, unflatten_WH, n_params = make_flat_funcs(L,
                                H_WH = jax.hessian(flat_action_WH)(theta_flat)
                                H_WH_np = np.array(H_WH, dtype=np.float64)
                                evals_WH = np.linalg.eigvalsh(H_WH_np)
                                lam_mins.append(float(evals_WH[0]))

            lam_mins = np.array(lam_mins)
            worst = lam_mins.min()
            results.append((beta, scale, worst))

    print(f"L={L}, c0={c0}, n_samples={n_samples}")

```

```

    print("beta    scale    min_lambda_min(W+Haar) over samples")
    for beta, scale, worst in results:
        print(f"{beta:4.1f}    {scale:5.3f}    {worst:+.6f}")

    return results

# ---- run convexity grid scan ----
L = 2
betas = jnp.linspace(0.4, 2.0, 9)    # 0.4,0.6,...,2.0
scales = (0.05, 0.10, 0.15, 0.20)
c0 = 0.25
n_samples = 4

convexity_grid = estimate_convexity_grid(L=L,
                                         betas=betas,
                                         scales=scales,
                                         c0=c0,
                                         n_samples=n_samples,
                                         seed=4243)

```

```

L=2, c0=0.25, n_samples=4
beta    scale    min_lambda_min(W+Haar) over samples
0.4    0.050    +0.221707
0.4    0.100    +0.192069
0.4    0.150    +0.155557
0.4    0.200    +0.128275
0.6    0.050    +0.208410
0.6    0.100    +0.162905
0.6    0.150    +0.113973
0.6    0.200    +0.050470
0.8    0.050    +0.193895
0.8    0.100    +0.135138
0.8    0.150    +0.077318
0.8    0.200    +0.005620
1.0    0.050    +0.181631
1.0    0.100    +0.110958
1.0    0.150    +0.016347
1.0    0.200    -0.083529
1.2    0.050    +0.167401
1.2    0.100    +0.067437
1.2    0.150    -0.015318
1.2    0.200    -0.099050
1.4    0.050    +0.152704
1.4    0.100    +0.039497
1.4    0.150    -0.073476
1.4    0.200    -0.183942
1.6    0.050    +0.140088
1.6    0.100    +0.043783
1.6    0.150    -0.088179
1.6    0.200    -0.219614
1.8    0.050    +0.132931
1.8    0.100    -0.022873
1.8    0.150    -0.116557

```

```

1.0  0.100  0.170332
1.8  0.200 -0.294094
2.0  0.050 +0.112468
2.0  0.100 -0.045922
2.0  0.150 -0.196407
2.0  0.200 -0.330255

```

```

# =====
# SU(3) WILSON + HAAR + LANCZOS (A100-OPTIMIZED, DIFFERENTIABLE)
# =====

import jax
import jax.numpy as jnp
import numpy as np
import jax.lax as lax

jax.config.update("jax_enable_x64", False)  # A100-friendly FP32

# =====
# 1. SU(3) BASIS
# =====
def su3_generators():
    lam1 = jnp.array([[0,1,0],[1,0,0],[0,0,0]], jnp.complex64)
    lam2 = jnp.array([[0,-1j,0],[1j,0,0],[0,0,0]], jnp.complex64)
    lam3 = jnp.array([[1,0,0],[0,-1,0],[0,0,0]], jnp.complex64)
    lam4 = jnp.array([[0,0,1],[0,0,0],[1,0,0]], jnp.complex64)
    lam5 = jnp.array([[0,0,-1j],[0,0,0],[1j,0,0]], jnp.complex64)
    lam6 = jnp.array([[0,0,0],[0,0,1],[0,1,0]], jnp.complex64)
    lam7 = jnp.array([[0,0,0],[0,0,-1j],[0,1j,0]], jnp.complex64)
    lam8 = jnp.array([[1,0,0],[0,1,0],[0,0,-2]], jnp.complex64) / jnp.sqrt(3)
    lam = jnp.stack([lam1,lam2,lam3,lam4,lam5,lam6,lam7,lam8], 0)
    return 1j * lam / 2.0

T_SU3 = su3_generators()

def su3_alg_from_vec(a):
    return jnp.einsum("...a,aij->...ij", a, T_SU3)

# =====
# 2. DIFFERENTIABLE SU(3) EXPONENTIAL (Padé [1/1])
# =====
def su3_exp_differentiable(A):
    """
    Fully differentiable Padé [1/1] exponential approximation:
     $\exp(A) \approx (I - A/2)^{-1} (I + A/2)$ 
    Exact for Lie algebras to second order.
    Stable for anti-Hermitian matrices.
    Very fast on A100.
    """

```

```

    I = jnp.eye(3, dtype=jnp.complex64)
    X = 0.5 * A
    return jnp.linalg.solve(I - X, I + X)

# =====
# 3. STATIC build_links
# =====
def build_links_factory(L):
    def build_links(params):
        flat = params.reshape(-1,8)
        A = jax.vmap(su3_alg_from_vec)(flat)
        U = jax.vmap(su3_exp_differentiable)(A) # <--- FIXED EXP
        return U.reshape(L,L,L,L,4,3,3)
    return build_links

# =====
# 4. VECTOR WILSON ACTION
# =====
def vectorized_wilson_action(params, L, beta, build_links_L):
    U = build_links_L(params)

    S = 0.0
    for mu in range(4):
        for nu in range(mu+1,4):
            U_mu = U[..., mu, :, :]
            U_nu_shift = jnp.roll(U[..., nu, :, :], -1, axis=mu)

            U_mu_dag_shift = jnp.transpose(
                jnp.conjugate(jnp.roll(U[..., mu, :, :], -1, axis=nu)),
                (0,1,2,3,5,4)
            )

            U_nu_dag = jnp.transpose(jnp.conjugate(U[..., nu, :, :]),
                                    (0,1,2,3,5,4))

            P = U_mu @ U_nu_shift @ U_mu_dag_shift @ U_nu_dag
            trP = jnp.real(jnp.einsum("...ii->...", P))
            S += jnp.sum(beta * (1 - trP/3))
    return S

# =====
# 5. HAAR MASS
# =====
def haar_mass(params, c0):
    flat = params.reshape(-1,8)
    def per(a):
        A = su3_alg_from_vec(a)
        return jnp.real(jnp.trace(A.conj().T @ A))

```

```

        return jnp.real(jnp.trace(A.conj()).sum())
    return c0 * jax.vmap(per)(flat).sum()

# =====
# 6. TOTAL ACTION (JIT-FUSED)
# =====
def make_flat_funcs(L, beta, c0):
    build_links_L = build_links_factory(L)

    def unflatten(x):
        return x.reshape((L,L,L,L,4,8))

    def flat_action(theta):
        params = unflatten(theta)
        return vectorized_wilson_action(params, L, beta, build_links_L) \
            + haar_mass(params, c0)

    return jax.jit(flat_action), unflatten, (L**4*4*8)

# =====
# 7. HVP
# =====
def hvp(flat_action, theta, v):
    g = jax.grad(flat_action)
    _, hv = jax.jvp(g, (theta,), (v,))
    return hv

# =====
# 8. LANCZOS (GPU-FUSED)
# =====
def lanczos_min(flat_action, theta, k=20, seed=0):
    key = jax.random.PRNGKey(seed)
    n = theta.shape[0]

    v0 = jax.random.normal(key, (n,))
    v0 = v0 / jnp.linalg.norm(v0)

    def step(carry, _):
        v_prev, v, beta_prev = carry

        w = hvp(flat_action, theta, v)
        w = w - beta_prev * v_prev

        alpha = jnp.dot(w, v)
        w = w - alpha * v

        beta = jnp.linalg.norm(w)
        v_next = w / (beta + 1e-8)

```

```

        return (v, v_next, beta), (alpha, beta)

(_, _, _), (alphas, betas) = lax.scan(
    step,
    init=(jnp.zeros_like(v0), v0, 0.0),
    xs=None,
    length=k
)

alphas = jnp.array(alphas)
betas = jnp.array(betas[:-1])
T = jnp.diag(alphas) + jnp.diag(betas,1) + jnp.diag(betas,-1)

return float(jnp.linalg.eigvalsh(T)[0])

# =====
# 9. RANDOM THETA
# =====
def sample_theta(L, scale, key):
    return (scale * jax.random.normal(key, (L,L,L,L,4,8))).reshape(-1)

# =====
# 10. CONVEXITY GRID
# =====
def convexity_grid(L, betas, scales, c0=0.25, n_samples=3, seed=1):
    key = jax.random.PRNGKey(seed)
    results = []

    for beta in betas:
        flat_action, unflatten, n_params = make_flat_funcs(L, float(beta), c

        for scale in scales:
            lam_vals = []
            for _ in range(n_samples):
                key, sub = jax.random.split(key)
                theta = sample_theta(L, scale, sub)

                key, sub2 = jax.random.split(key)
                lam = lanczos_min(flat_action, theta, k=20, seed=int(sub2[0])
                lam_vals.append(lam)

            results.append((float(beta), scale, float(np.min(lam_vals))))

    print(f"\n=== L={L}, n_params={n_params} ===")
    print("beta    scale    lambda_min")
    for beta, scale, lam in results:
        print(f"{beta:4.1f}    {scale:5.3f}    {lam:+.6f}")

```

```

    return results

# =====
# RUN (A100)
# =====
betas = jnp.linspace(0.4, 2.0, 9)
scales = (0.05, 0.10, 0.15)

print("Running SU(3) convexity scan for L = 3 on A100 ...")
res = convexity_grid(3, betas, scales, c0=0.25, n_samples=3)

print("\nDone.")

```

```

Running SU(3) convexity scan for L = 3 on A100 ...
/usr/local/lib/python3.12/dist-packages/jax/_src/lax/lax.py:5473: ComplexWarning
  x_bar = _convert_element_type(x_bar, x.aval.dtype, x.aval.weak_type)

```

```

=== L=3, n_params=2592 ===
beta    scale    lambda_min
0.4    0.050    +0.229415
0.4    0.100    +0.205332
0.4    0.150    +0.182681
0.6    0.050    +0.220478
0.6    0.100    +0.184114
0.6    0.150    +0.142536
0.8    0.050    +0.211304
0.8    0.100    +0.164696
0.8    0.150    +0.105439
1.0    0.050    +0.200962
1.0    0.100    +0.142384
1.0    0.150    +0.078616
1.2    0.050    +0.191056
1.2    0.100    +0.120973
1.2    0.150    +0.041559
1.4    0.050    +0.180576
1.4    0.100    +0.101330
1.4    0.150    +0.004239
1.6    0.050    +0.171409
1.6    0.100    +0.079299
1.6    0.150    -0.029265
1.8    0.050    +0.157365
1.8    0.100    +0.056300
1.8    0.150    -0.063324
2.0    0.050    +0.145479
2.0    0.100    +0.025839
2.0    0.150    -0.098911

```

```
Done.
```

```

# =====
# RESEARCH-GRADE SU(3) CONVEXITY ENGINE

```

```

# A100-optimized, mathematically consistent, fully differentiable
#
# Components:
#   • SU(3) algebra map and correct Haar curvature term ( $N/24 = 1/8$ )
#   • Stable Padé [2/2] exponential for  $||A|| < 1$ 
#   • Differentiable Wilson action with correct plaquette geometry
#   • JAX checkpointing for memory-efficient Hessian-vector products
#   • Lanczos extremal eigenvalue solver ( $\lambda_{\min}$ )
#   • Unitarity diagnostics ( $||U^\dagger U - I||$ )
#   • Static shape specialization for stable XLA compilation
#
# Supports  $L = 2, 3, 4, 6$ , and  $L = 8$  with patience.
# =====

import jax
import jax.numpy as jnp
import numpy as np
import jax.lax as lax
import time

jax.config.update("jax_enable_x64", False) # FP32 sufficient for curvature

# =====
# 0. RESEARCH CONSTANTS
# =====
# Haar quadratic coefficient ( $N/24$ ) for SU(N)
C0_SU3 = 3.0 / 24.0 # = 0.125

# Stability region for Padé[2/2]:
#   Safe for  $||A|| \leq 1.0$ 
#   Link unitarity error:  $O(||A||^6)$ 
PADE22_MAX_NORM = 1.0

# =====
# 1. SU(3) GENERATORS (Gell-Mann basis)
# =====
def su3_generators():
    """Return the standard anti-Hermitian SU(3) generators  $T_a = i \lambda_a / 2$ ."""
    lam1 = jnp.array([[0, 1, 0], [1, 0, 0], [0, 0, 0]], jnp.complex64)
    lam2 = jnp.array([[0, -1j, 0], [1j, 0, 0], [0, 0, 0]], jnp.complex64)
    lam3 = jnp.array([[1, 0, 0], [0, -1, 0], [0, 0, 0]], jnp.complex64)
    lam4 = jnp.array([[0, 0, 1], [0, 0, 0], [1, 0, 0]], jnp.complex64)
    lam5 = jnp.array([[0, 0, -1j], [0, 0, 0], [1j, 0, 0]], jnp.complex64)
    lam6 = jnp.array([[0, 0, 0], [0, 0, 1], [0, 1, 0]], jnp.complex64)
    lam7 = jnp.array([[0, 0, 0], [0, 0, -1j], [0, 1j, 0]], jnp.complex64)
    lam8 = jnp.array([[1, 0, 0], [0, 1, 0], [0, 0, -2]], jnp.complex64) / jnp.sqrt(3)

    lam = jnp.stack([lam1, lam2, lam3, lam4, lam5, lam6, lam7, lam8], axis=0)

```

```

        return 1j * lam / 2.0

T_SU3 = su3_generators()

def su3_alg_from_vec(a):
    """Map  $\mathbb{R}^8 \rightarrow \text{su}(3)$  matrix using generators."""
    return jnp.einsum("...a,aij->...ij", a, T_SU3)

# =====
# 2. SU(3) EXPONENTIAL VIA PADÉ [2/2]
# =====
@jax.checkpoint
def su3_exp_pade22(A):
    """
    Differentiable Padé approximant  $\exp(A)$ 
    -----
     $\exp(A) \approx (I - A/2 + A^2/12)^{-1} (I + A/2 + A^2/12)$ 

    Properties:
    • Accurate to  $O(A^5)$ 
    • Stable for  $||A|| \leq 1$ 
    • Preserves approximate unitarity for  $\text{su}(3)$ 
    • Fully JAX differentiable (no eigenvectors)

    Suitable for curvature experiments where  $||A|| \sim O(0.1-0.5)$ .
    """
    I = jnp.eye(3, dtype=jnp.complex64)
    A2 = A @ A
    c1 = 0.5
    c2 = 1/12.0

    Num = I + c1*A + c2*A2
    Den = I - c1*A + c2*A2

    return jnp.linalg.solve(Den, Num)

# =====
# 3. STATIC-LINK BUILDER
# =====
def build_links_factory(L):
    @jax.checkpoint
    def build_links(params):
        """ $\theta \rightarrow U$  (differentiable, rematerialized during backward pass)"""
        flat = params.reshape(-1,8)
        A = jax.vmap(su3_alg_from_vec)(flat)
        U = jax.vmap(su3_exp_pade22)(A)
        return U.reshape(L,L,L,L,4,3,3)
    return build_links

```

```

# =====
# 4. VECTORIZED WILSON ACTION (Fully Differentiable)
# =====
@jax.checkpoint
def compute_plaquette_sum(U, beta):
    """
    Sum all oriented plaquettes:
     $S_W = \beta \sum_{\{x, \mu < \nu\}} [1 - (1/3) \text{Re Tr}(U_\mu U_\nu(x+\mu) U_\mu^\dagger(x+\nu) U_\nu^\dagger)]$ 
    """
    S = 0.0
    for mu in range(4):
        for nu in range(mu+1,4):
            U_mu = U[..., mu, :, :]
            U_nu_shift = jnp.roll(U[..., nu, :, :], -1, axis=mu)

            U_mu_dag_shift = jnp.swapaxes(
                jnp.conjugate(jnp.roll(U[..., mu, :, :], -1, axis=nu)),
                -1, -2
            )
            U_nu_dag = jnp.swapaxes(jnp.conjugate(U[..., nu, :, :]), -1, -2)

            P = U_mu @ U_nu_shift @ U_mu_dag_shift @ U_nu_dag
            trP = jnp.real(jnp.einsum("...ii->...", P))

            S += jnp.sum(1.0 - trP/3.0)
    return beta * S

def vectorized_wilson_action(params, L, beta, build_links_L):
    """params → U → Wilson action"""
    U = build_links_L(params)
    return compute_plaquette_sum(U, beta)

# =====
# 5. HAAR MASS (Quadratic Approximation)
# =====
def haar_mass(params, c0):
    """
    SU(3) Haar measure local expansion:
     $\log J(\exp A) \approx c0 * \text{Tr}(A^2)$ 
    with  $c0 = N/24 = 0.125$  for SU(3).
    """
    flat = params.reshape(-1,8)

    def per(a):
        A = su3_alg_from_vec(a)
        return jnp.real(jnp.trace(A.conj().T @ A))

```

```

        return c0 * jax.vmap(per)(flat).sum()

# =====
# 6. TOTAL ACTION
# =====
def make_flat_funcs(L, beta, c0=C0_SU3):
    build_links_L = build_links_factory(L)

    def unflatten(theta):
        return theta.reshape((L,L,L,L,4,8))

    def flat_action(theta):
        params = unflatten(theta)
        S_W = vectorized_wilson_action(params, L, beta, build_links_L)
        S_H = haar_mass(params, c0)
        return S_W + S_H

    return jax.jit(flat_action), (L**4 * 4 * 8)

# =====
# 7. HESSIAN-VECTOR PRODUCT (HVP)
# =====
def hvp(flat_action, theta, v):
    """Return H v using JVP of grad(flat_action)."""
    g = jax.grad(flat_action)
    _, hv = jax.jvp(g, (theta,), (v,))
    return hv

# =====
# 8. LANCZOS EXTREMAL EIGENSOLVER
# =====
def lanczos_min(flat_action, theta, k=20, seed=0):
    """
    Estimate  $\lambda_{\min}$ (Hessian) via Lanczos applied to HVP routine.
    """
    key = jax.random.PRNGKey(seed)
    n = theta.shape[0]

    v0 = jax.random.normal(key, (n,))
    v0 /= jnp.linalg.norm(v0)

    def step(carry, _):
        v_prev, v, beta_prev = carry

        w = hvp(flat_action, theta, v)
        w -= beta_prev * v_prev

        alpha = jnp.dot(w, v)

```

```

        w -= alpha * v

    beta = jnp.linalg.norm(w)
    v_next = w / (beta + 1e-8)

    return (v, v_next, beta), (alpha, beta)

(_, _, _), (a, b) = lax.scan(
    step,
    init=(jnp.zeros_like(v0), v0, 0.0),
    xs=None,
    length=k,
)

a = jnp.array(a)
b = jnp.array(b[:-1])
T = jnp.diag(a) + jnp.diag(b,1) + jnp.diag(b,-1)

eigs = jnp.linalg.eigvalsh(T)
return float(eigs[0])

# =====
# 9. SAMPLING AND GRID SCAN
# =====
def sample_theta(L, scale, key):
    return (scale * jax.random.normal(key, (L,L,L,L,4,8))).reshape(-1)

def convexity_grid(L, betas, scales, c0=C0_SU3, n_samples=3, seed=1):
    """
    Compute  $\lambda_{\min}(\text{flat\_action}(\theta))$  over a grid of  $(\beta, \text{scale})$ 
    and report worst-case curvature.
    """
    key = jax.random.PRNGKey(seed)
    results = []

    for beta in betas:
        flat_action, n_params = make_flat_funcs(L, float(beta), c0)

        for scale in scales:
            lam_vals = []

            for _ in range(n_samples):
                key, sub = jax.random.split(key)
                theta = sample_theta(L, scale, sub)

                key, sub2 = jax.random.split(key)
                lam = lanczos_min(flat_action, theta, k=20, seed=int(sub2[0])
                lam_vals.append(lam)

```

```

        results.append((float(beta), scale, float(np.min(lam_vals))))

    print(f"\n=== L={L}, Haar c0={c0}, n_params={n_params} ===")
    print(" beta      scale      lambda_min")
    for b, sc, lam in results:
        print(f" {b:4.2f}      {sc:5.3f}      {lam:+.6f}")

    return results

# =====
# EXECUTION BLOCK (DEFAULT: L = 4)
# =====
betas = jnp.linspace(0.4, 3.0, 8)
scales = (0.05, 0.10, 0.15)

print("Running SU(3) convexity scan (Padé[2/2], Correct Haar, L=4)...")
t0 = time.time()

res = convexity_grid(4, betas, scales, n_samples=3)

print("\nCompleted in", time.time() - t0, "seconds.")

```

Running SU(3) convexity scan (Padé[2/2], Correct Haar, L=4)...

```

=== L=4, Haar c0=0.125, n_params=8192 ===
 beta      scale      lambda_min
0.40      0.050      +0.107639
0.40      0.100      +0.084942
0.40      0.150      +0.060163
0.77      0.050      +0.090999
0.77      0.100      +0.049703
0.77      0.150      +0.000575
1.14      0.050      +0.074027
1.14      0.100      +0.011488
1.14      0.150      -0.063704
1.51      0.050      +0.058620
1.51      0.100      -0.028256
1.51      0.150      -0.121915
1.89      0.050      +0.042761
1.89      0.100      -0.061317
1.89      0.150      -0.172886
2.26      0.050      +0.024951
2.26      0.100      -0.097842
2.26      0.150      -0.229981
2.63      0.050      +0.006105
2.63      0.100      -0.131974
2.63      0.150      -0.287083
3.00      0.050      -0.008208
3.00      0.100      -0.172180
3.00      0.150      -0.376565

```

Completed in 104.02800524072876 seconds

Completed in 194.92809554072870 seconds.

```

# =====
# FAST SU(3) CONVEXITY ENGINE
# A100-optimized • Correct Haar • Padé[2/2] • Proper Checkpointing
# =====

import jax
import jax.numpy as jnp
import numpy as np
import jax.lax as lax
import time

jax.config.update("jax_enable_x64", False) # A100: FP32 = optimal

# =====
# CONSTANTS
# =====
C0_SU3 = 3.0 / 24.0 # Correct Haar coefficient = 0.125

# =====
# 1. SU(3) GENERATORS
# =====
def su3_generators():
    lam1 = jnp.array([[0,1,0],[1,0,0],[0,0,0]], jnp.complex64)
    lam2 = jnp.array([[0,-1j,0],[1j,0,0],[0,0,0]], jnp.complex64)
    lam3 = jnp.array([[1,0,0],[0,-1,0],[0,0,0]], jnp.complex64)
    lam4 = jnp.array([[0,0,1],[0,0,0],[1,0,0]], jnp.complex64)
    lam5 = jnp.array([[0,0,-1j],[0,0,0],[1j,0,0]], jnp.complex64)
    lam6 = jnp.array([[0,0,0],[0,0,1],[0,1,0]], jnp.complex64)
    lam7 = jnp.array([[0,0,0],[0,0,-1j],[0,1j,0]], jnp.complex64)
    lam8 = jnp.array([[1,0,0],[0,1,0],[0,0,-2]], jnp.complex64) / jnp.sqrt(3)

    lam = jnp.stack([lam1,lam2,lam3,lam4,lam5,lam6,lam7,lam8], 0)
    return 1j * lam / 2.0

T_SU3 = su3_generators()

def su3_alg_from_vec(a):
    return jnp.einsum("...a,aij->...ij", a, T_SU3)

# =====
# 2. SU(3) EXPONENTIAL VIA STABLE PADÉ [2/2] (CHECKPOINTED)
# =====
@jax.checkpoint
def su3_exp_pade22(A):
    """
    Padé [2/2] approximant:

```

```

    raise ValueError("approximate.")
    exp(A) ≈ (I - A/2 + A^2/12)^(-1)(I + A/2 + A^2/12)
    Accurate and stable for ||A|| < ~1.
    Fully differentiable.
    """
    I = jnp.eye(3, dtype=jnp.complex64)
    A2 = A @ A

    c1 = 0.5
    c2 = 1/12.0

    Num = I + c1*A + c2*A2
    Den = I - c1*A + c2*A2

    return jnp.linalg.solve(Den, Num)

# =====
# 3. BUILD LINKS (CHECKPOINTED)
# =====
def build_links_factory(L):

    @jax.checkpoint
    def build_links(params):
        flat = params.reshape(-1,8)
        A = jax.vmap(su3_alg_from_vec)(flat)
        U = jax.vmap(su3_exp_pade22)(A)
        return U.reshape(L,L,L,L,4,3,3)

    return build_links

# =====
# 4. WILSON PLAQUETTE SUM (NO CHECKPOINT HERE – FOR SPEED)
# =====
def compute_plaquette_sum(U, beta):
    S = 0.0

    for mu in range(4):
        for nu in range(mu+1,4):
            U_mu = U[..., mu, :, :]
            U_nu_shift = jnp.roll(U[..., nu, :, :], -1, axis=mu)

            U_mu_dag_shift = jnp.swapaxes(
                jnp.conjugate(jnp.roll(U[..., mu, :, :], -1, axis=nu)),
                -1, -2
            )
            U_nu_dag = jnp.swapaxes(jnp.conjugate(U[..., nu, :, :]), -1, -2)

            P = U_mu @ U_nu_shift @ U_mu_dag_shift @ U_nu_dag

```

```

        trP = jnp.real(jnp.einsum("...ii->...", P))
        S += jnp.sum(1.0 - trP/3.0)

    return beta * S

def vectorized_wilson_action(params, L, beta, build_links_L):
    U = build_links_L(params)
    return compute_plaquette_sum(U, beta)

# =====
# 5. HAAR MASS (Quadratic SU(3) Approx.)
# =====
def haar_mass(params, c0):
    flat = params.reshape(-1,8)

    def per(a):
        A = su3_alg_from_vec(a)
        return jnp.real(jnp.trace(A.conj().T @ A))

    return c0 * jax.vmap(per)(flat).sum()

# =====
# 6. TOTAL ACTION
# =====
def make_flat_funcs(L, beta, c0=C0_SU3):

    build_links_L = build_links_factory(L)

    def unflatten(theta):
        return theta.reshape((L,L,L,L,4,8))

    def flat_action(theta):
        params = unflatten(theta)
        return (
            vectorized_wilson_action(params, L, beta, build_links_L)
            + haar_mass(params, c0)
        )

    return jax.jit(flat_action), (L**4 * 4 * 8)

# =====
# 7. HVP & LANCZOS
# =====
def hvp(flat_action, theta, v):
    g = jax.grad(flat_action)
    _, hv = jax.jvp(g, (theta,), (v,))
    return hv

```

```

    return iv

def lanczos_min(flat_action, theta, k=20, seed=0):
    key = jax.random.PRNGKey(seed)
    n = theta.shape[0]

    v0 = jax.random.normal(key, (n,))
    v0 /= jnp.linalg.norm(v0)

    def step(carry, _):
        v_prev, v, beta_prev = carry

        w = hvp(flat_action, theta, v)
        w -= beta_prev * v_prev

        alpha = jnp.dot(w, v)
        w -= alpha * v

        beta = jnp.linalg.norm(w)
        v_next = w / (beta + 1e-8)

        return (v, v_next, beta), (alpha, beta)

    (_, _, _), (a, b) = lax.scan(step,
                                   init=(jnp.zeros_like(v0), v0, 0.0),
                                   xs=None,
                                   length=k)

    a = jnp.array(a)
    b = jnp.array(b[:-1])
    T = jnp.diag(a) + jnp.diag(b,1) + jnp.diag(b,-1)

    return float(jnp.linalg.eigvalsh(T)[0])

# =====
# 8. ( $\beta$ , scale) CONVEXITY GRID
# =====
def sample_theta(L, scale, key):
    return (scale * jax.random.normal(key, (L,L,L,L,4,8))).reshape(-1)

def convexity_grid(L, betas, scales, c0=C0_SU3, n_samples=3, seed=1):
    key = jax.random.PRNGKey(seed)
    results = []

    for beta in betas:
        flat_action, n_params = make_flat_funcs(L, float(beta), c0)

        for scale in scales:
            lam_vals = []

```

```

        for _ in range(n_samples):
            key, sub = jax.random.split(key)
            theta = sample_theta(L, scale, sub)

            key, sub2 = jax.random.split(key)
            lam = lanczos_min(flat_action, theta, k=20, seed=int(sub2[0])
            lam_vals.append(lam)

        results.append((float(beta), scale, float(np.min(lam_vals))))

    print(f"\n=== L={L}, Haar={c0}, params={n_params} ===")
    print(" beta      scale      lambda_min")
    for b, sc, lam in results:
        print(f" {b:4.2f}      {sc:5.3f}      {lam:+.6f}")

    return results

# =====
# RUN L = 4 (FAST)
# =====
betas = jnp.linspace(0.4, 3.0, 8)
scales = (0.05, 0.10, 0.15)

print("Running FAST SU(3) convexity scan (L=4)...")
t0 = time.time()

res = convexity_grid(4, betas, scales)

print("\nCompleted in", time.time() - t0, "seconds.")

```

Running FAST SU(3) convexity scan (L=4)...

=== L=4, Haar=0.125, params=8192 ===

beta	scale	lambda_min
0.40	0.050	+0.107639
0.40	0.100	+0.084942
0.40	0.150	+0.060163
0.77	0.050	+0.090999
0.77	0.100	+0.049703
0.77	0.150	+0.000575
1.14	0.050	+0.074027
1.14	0.100	+0.011488
1.14	0.150	-0.063704
1.51	0.050	+0.058620
1.51	0.100	-0.028256
1.51	0.150	-0.121915
1.89	0.050	+0.042761
1.89	0.100	-0.061317
1.89	0.150	-0.172886

2.26	0.050	+0.024951
2.26	0.100	-0.097842
2.26	0.150	-0.229981
2.63	0.050	+0.006105
2.63	0.100	-0.131974
2.63	0.150	-0.287083
3.00	0.050	-0.008208
3.00	0.100	-0.172180
3.00	0.150	-0.376565

Completed in 180.61275458335876 seconds.

```
# =====
# TURBO SU(3) CONVEXITY ENGINE
# np.conjugate(jnp.roll(U[..., mu, :, :], -1, axis=nu)), -1, -2)
    U_nu_dag = jnp.swapaxes(jnp.conjugate(U[..., nu, :, :]), -1, -2)

    P = U_mu @ U_nu_shift @ U_mu_dag_shift @ U_nu_dag
    trP = jnp.real(jnp.einsum("...ii->...", P))
    S += jnp.sum(1.0 - trP/3.0)
return beta * S

def vectorized_wilson_action(params, L, beta, build_links_L):
    U = build_links_L(params)
    return compute_plaquette_sum(U, beta)

# =====
# 5. HAAR MASS (Correct c0=0.125)
# =====
def haar_mass(params, c0):
    flat = params.reshape(-1,8)
    def per(a):
        A = su3_alg_from_vec(a)
        return jnp.real(jnp.trace(A.conj().T @ A))
    return c0 * jax.vmap(per)(flat).sum()

# =====
# 6. TOTAL ACTION & HVP
# =====
def make_flat_funcs(L, beta, c0):
    build_links_L = build_links_factory(L)
    def unflatten(theta): return theta.reshape((L,L,L,L,4,8))
    def flat_action(theta):
        params = unflatten(theta)
        return vectorized_wilson_action(params, L, beta, build_links_L) + ha
    return jax.jit(flat_action), (L**4 * 4 * 8)

def hvp(flat_action, theta, v):
    g = jax.grad(flat_action)
    _, hv = jax.jvp(g, (theta,), (v,))
    return hv
```

```

def lanczos_min(flat_action, theta, k=20, seed=0):
    key = jax.random.PRNGKey(seed)
    n = theta.shape[0]
    v0 = jax.random.normal(key, (n,))
    v0 /= jnp.linalg.norm(v0)

    def step(carry, _):
        v_prev, v, beta_prev = carry
        w = hvp(flat_action, theta, v)
        w -= beta_prev * v_prev
        alpha = jnp.dot(w, v)
        w -= alpha * v
        beta = jnp.linalg.norm(w)
        v_next = w / (beta + 1e-8)
        return (v, v_next, beta), (alpha, beta)

    (_, _, _), (a, b) = lax.scan(step, init=(jnp.zeros_like(v0), v0, 0.0), x
a = jnp.array(a); b = jnp.array(b[:-1])
T = jnp.diag(a) + jnp.diag(b,1) + jnp.diag(b,-1)
return float(jnp.linalg.eigvalsh(T)[0])

# =====
# 7. FAST SCANNER
# =====
def convexity_grid(L, betas, scales, c0=0.125, n_samples=3, seed=1):
    key = jax.random.PRNGKey(seed)
    print(f"\n--- Turbo Scan L={L}, Haar={c0} ---")

    start_time = time.time()
    for beta in betas:
        flat_action, _ = make_flat_funcs(L, float(beta), c0)
        for scale in scales:
            vals = []
            for _ in range(n_samples):
                key, sub = jax.random.split(key); key, sub2 = jax.random.spl
                theta = (scale * jax.random.normal(sub, (L,L,L,L,4,8))).resh
                vals.append(lanczos_min(flat_action, theta, k=20, seed=int(s

            min_lam = np.min(vals)
            print(f"beta={beta:4.2f} scale={scale:5.3f} lam={min_lam:+.6f}")

    print(f"Total Time: {time.time()-start_time:.2f}s")

# =====
# RUN
# =====
betas = jnp.linspace(0.4, 3.0, 8)
scales = (0.05, 0.10, 0.15)
convexity_grid(4, betas, scales)Fixed Checkpointing Strategy: Faster Executi
# =====

```

```

import jax
import jax.numpy as jnp
import numpy as np
import jax.lax as lax
import time

jax.config.update("jax_enable_x64", False) # A100: FP32 optimal

# =====
# 1. SU(3) GENERATORS & ALGEBRA
# =====
def su3_generators():
    lam1 = jnp.array([[0,1,0],[1,0,0],[0,0,0]], jnp.complex64)
    lam2 = jnp.array([[0,-1j,0],[1j,0,0],[0,0,0]], jnp.complex64)
    lam3 = jnp.array([[1,0,0],[0,-1,0],[0,0,0]], jnp.complex64)
    lam4 = jnp.array([[0,0,1],[0,0,0],[1,0,0]], jnp.complex64)
    lam5 = jnp.array([[0,0,-1j],[0,0,0],[1j,0,0]], jnp.complex64)
    lam6 = jnp.array([[0,0,0],[0,0,1],[0,1,0]], jnp.complex64)
    lam7 = jnp.array([[0,0,0],[0,0,-1j],[0,1j,0]], jnp.complex64)
    lam8 = jnp.array([[1,0,0],[0,1,0],[0,0,-2]], jnp.complex64) / jnp.sqrt(3)
    return 1j * jnp.stack([lam1,lam2,lam3,lam4,lam5,lam6,lam7,lam8], 0) / 2.

T_SU3 = su3_generators()

def su3_alg_from_vec(a):
    return jnp.einsum("...a,aij->...ij", a, T_SU3)

# =====
# 2. SU(3) EXPONENTIAL (KEEP CHECKPOINT)
# =====
# We checkpoint this because storing 'A' is cheaper than storing 'U'
@jax.checkpoint
def su3_exp_pade22(A):
    I = jnp.eye(3, dtype=jnp.complex64)
    A2 = A @ A
    c1, c2 = 0.5, 0.0833333336
    Num = I + c1*A + c2*A2
    Den = I - c1*A + c2*A2
    return jnp.linalg.solve(Den, Num)

# =====
# 3. BUILD LINKS (KEEP CHECKPOINT)
# =====
def build_links_factory(L):
    @jax.checkpoint
    def build_links(params):
        flat = params.reshape(-1,8)
        A = jax.vmap(su3_alg_from_vec)(flat)
        U = jax.vmap(su3_exp_pade22)(A)
        . . .

```

```

        return U.reshape(L,L,L,L,4,3,3)
    return build_links

# =====
# 4. WILSON ACTION (REMOVE CHECKPOINT!!!)
# =====
# FIX: No @jax.checkpoint here. We want JAX to store the intermediate
# graph for speed, because recomputing the matrix muls is too expensive.
def compute_plaquette_sum(U, beta):
    S = 0.0
    for mu in range(4):
        for nu in range(mu+1,4):
            U_mu = U[..., mu, :, :]
            U_nu_shift = jnp.roll(U[..., nu, :, :], -1, axis=mu)
            U_mu_dag_shift = jnp.swapaxes(j

```

```

--- Turbo Scan L=4, Haar=0.125 ---
beta=0.40 scale=0.050 lam=+0.107639
beta=0.40 scale=0.100 lam=+0.084942
beta=0.40 scale=0.150 lam=+0.060163
beta=0.77 scale=0.050 lam=+0.090999
beta=0.77 scale=0.100 lam=+0.049703
beta=0.77 scale=0.150 lam=+0.000575
beta=1.14 scale=0.050 lam=+0.074027
beta=1.14 scale=0.100 lam=+0.011488
beta=1.14 scale=0.150 lam=-0.063704
beta=1.51 scale=0.050 lam=+0.058620
beta=1.51 scale=0.100 lam=-0.028256
beta=1.51 scale=0.150 lam=-0.121915
beta=1.89 scale=0.050 lam=+0.042761
beta=1.89 scale=0.100 lam=-0.061317
beta=1.89 scale=0.150 lam=-0.172886
beta=2.26 scale=0.050 lam=+0.024951
beta=2.26 scale=0.100 lam=-0.097842
beta=2.26 scale=0.150 lam=-0.229981
beta=2.63 scale=0.050 lam=+0.006105
beta=2.63 scale=0.100 lam=-0.131974
beta=2.63 scale=0.150 lam=-0.287083
beta=3.00 scale=0.050 lam=-0.008208
beta=3.00 scale=0.100 lam=-0.172180
beta=3.00 scale=0.150 lam=-0.376565
Total Time: 179.41s

```

```

# =====
# RUN L = 6 ONLY (Next scientific step)
# =====

betas = jnp.linspace(0.4, 3.0, 8)
scales = (0.05, 0.10, 0.15)

print("\n### Running convexity scan for L = 6 ###")
res6 = convexity_grid(6, betas, scales)

```

```
print("\nL = 6 scan complete.")
```

```
### Running convexity scan for L = 6 ###
```

```
--- JIT-ISOLATED Scan L=6 (Expect ~20s) ---
```

```
beta=0.40 scale=0.050 lam=+0.108966
beta=0.40 scale=0.100 lam=+0.087381
beta=0.40 scale=0.150 lam=+0.063117
beta=0.77 scale=0.050 lam=+0.093839
beta=0.77 scale=0.100 lam=+0.052703
beta=0.77 scale=0.150 lam=+0.006658
beta=1.14 scale=0.050 lam=+0.079105
beta=1.14 scale=0.100 lam=+0.016544
beta=1.14 scale=0.150 lam=-0.052778
beta=1.51 scale=0.050 lam=+0.063542
beta=1.51 scale=0.100 lam=-0.017121
beta=1.51 scale=0.150 lam=-0.111918
beta=1.89 scale=0.050 lam=+0.048837
beta=1.89 scale=0.100 lam=-0.056489
beta=1.89 scale=0.150 lam=-0.173765
beta=2.26 scale=0.050 lam=+0.033850
beta=2.26 scale=0.100 lam=-0.085747
beta=2.26 scale=0.150 lam=-0.232562
beta=2.63 scale=0.050 lam=+0.018730
beta=2.63 scale=0.100 lam=-0.120620
beta=2.63 scale=0.150 lam=-0.278895
beta=3.00 scale=0.050 lam=+0.003391
beta=3.00 scale=0.100 lam=-0.154899
beta=3.00 scale=0.150 lam=-0.348172
Total Time: 193.88s
```

```
L = 6 scan complete.
```

```
# =====
# RUN L = 8 - Minimal Scientific Scan
#   • Only scale=0.05 (small-field convexity region)
#   • Full beta sweep (0.4 → 3.0)
#   • n_samples = 2 (reduces runtime / compute)
#   • Expected A100 runtime: 5-7 minutes
#   • Key Question: Does convexity persist at large volume?
# =====
```

```
betas = jnp.linspace(0.4, 3.0, 8)
scales = (0.05,)      # <-- minimal scan, scientifically relevant
n_samples = 2         # <-- reduces compute without harming signal
```

```
print("\n### Running convexity scan for L = 8 (minimal slice) ###")
t0 = time.time()
```

```
res8 = convexity_grid(8, betas, scales, 60, 0.125, n_samples, n_samples)
```

```

res8 = convexity_grid(8, betas, scales, c0=0.125, n_samples=n_samples)

print("\nL = 8 minimal scan complete.")
print("Total Time:", time.time() - t0, "seconds")

# Optionally store the L=8 results:
res8

```

```

### Running convexity scan for L = 8 (minimal slice) ###

```

```

--- JIT-ISOLATED Scan L=8 (Expect ~20s) ---

```

```

beta=0.40 scale=0.050 lam=+0.109223
beta=0.77 scale=0.050 lam=+0.094255
beta=1.14 scale=0.050 lam=+0.079963
beta=1.51 scale=0.050 lam=+0.064737
beta=1.89 scale=0.050 lam=+0.050365
beta=2.26 scale=0.050 lam=+0.035428
beta=2.63 scale=0.050 lam=+0.021559
beta=3.00 scale=0.050 lam=+0.006509
Total Time: 41.65s

```

```

L = 8 minimal scan complete.
Total Time: 41.65427279472351 seconds

```

```

# =====
# RECONSTRUCTED convexity results (from your logged output)
# No recomputation required.
# =====

```

```

res4 = [
    (0.40, 0.050, +0.107639),
    (0.40, 0.100, +0.084942),
    (0.40, 0.150, +0.060163),

    (0.77, 0.050, +0.090999),
    (0.77, 0.100, +0.049703),
    (0.77, 0.150, +0.000575),

    (1.14, 0.050, +0.074027),
    (1.14, 0.100, +0.011488),
    (1.14, 0.150, -0.063704),

    (1.51, 0.050, +0.058620),
    (1.51, 0.100, -0.028256),
    (1.51, 0.150, -0.121915),

    (1.89, 0.050, +0.042761),
    (1.89, 0.100, -0.061317),
    (1.89, 0.150, -0.172886),

```

```

(2.26, 0.050, +0.024951),
(2.26, 0.100, -0.097842),
(2.26, 0.150, -0.229981),

(2.63, 0.050, +0.006105),
(2.63, 0.100, -0.131974),
(2.63, 0.150, -0.287083),

(3.00, 0.050, -0.008208),
(3.00, 0.100, -0.172180),
(3.00, 0.150, -0.376565),
]

```

```

res6 = [
    (0.40, 0.050, +0.108966),
    (0.40, 0.100, +0.087381),
    (0.40, 0.150, +0.063117),

    (0.77, 0.050, +0.093839),
    (0.77, 0.100, +0.052703),
    (0.77, 0.150, +0.006658),

    (1.14, 0.050, +0.079105),
    (1.14, 0.100, +0.016544),
    (1.14, 0.150, -0.052778),

    (1.51, 0.050, +0.063542),
    (1.51, 0.100, -0.017121),
    (1.51, 0.150, -0.111918),

    (1.89, 0.050, +0.048837),
    (1.89, 0.100, -0.056489),
    (1.89, 0.150, -0.173765),

    (2.26, 0.050, +0.033850),
    (2.26, 0.100, -0.085747),
    (2.26, 0.150, -0.232562),

    (2.63, 0.050, +0.018730),
    (2.63, 0.100, -0.120620),
    (2.63, 0.150, -0.278895),

    (3.00, 0.050, +0.003391),
    (3.00, 0.100, -0.154899),
    (3.00, 0.150, -0.348172),
]

```

```

res8 = [
    (0.40, 0.050, +0.109223),
    (0.77, 0.050, +0.094255),
    (1.14, 0.050, +0.079105),

```

```
(1.14, 0.050, +0.079963),  
(1.51, 0.050, +0.064737),  
(1.89, 0.050, +0.050365),  
(2.26, 0.050, +0.035428),  
(2.63, 0.050, +0.021559),  
(3.00, 0.050, +0.006509),  
]  
  
print("Reconstruction complete:")  
print("res4:", len(res4), "entries")  
print("res6:", len(res6), "entries")  
print("res8:", len(res8), "entries")
```

```
Reconstruction complete:  
res4: 24 entries  
res6: 24 entries  
res8: 8 entries
```

```
res8_only = res8    # rename for clarity  
  
betas8, lam8 = zip(*[(b, lam) for (b, sc, lam) in res8_only])  
plt.plot(betas8, lam8, marker='o')  
plt.axhline(0, color='black', linewidth=1)  
plt.xlabel("beta")  
plt.ylabel("lambda_min")  
plt.title("Convexity: L=8, scale=0.05")  
plt.grid(True)  
plt.show()
```

```
# =====
# COMBINED CONVEXITY PLOT: L = 4, L = 6, L = 8 (scale = 0.05)
# =====

import matplotlib.pyplot as plt
import numpy as np

def extract_curve(res, target_scale=0.05):
    """Convert result list into sorted beta and lambda_min arrays."""
    if res is None:
        return None
    pts = [(b, lam) for (b, sc, lam) in res if abs(sc - target_scale) < 1e-1]
    if len(pts) == 0:
        return None
    pts = sorted(pts) # sort by beta
    betas = np.array([p[0] for p in pts])
    lambdas = np.array([p[1] for p in pts])
    return betas, lambdas

curves = {}

# Safely extract curves if variables exist
if 'res4' in globals():
    c4 = extract_curve(res4)
    if c4: curves['L=4'] = c4

if 'res6' in globals():
    c6 = extract_curve(res6)
    if c6: curves['L=6'] = c6

if 'res8' in globals():
    c8 = extract_curve(res8)
    if c8: curves['L=8'] = c8

# Check we have something to plot
if len(curves) == 0:
    raise RuntimeError("No convexity results found. Make sure res4/res6/res8

# ----- PLOT -----
plt.figure(figsize=(9,6))

for label, (bvals, lvals) in curves.items():
    plt.plot(bvals, lvals, marker='o', linewidth=2, label=label)

plt.axhline(0.0, color='black', linewidth=1)
```

```
plt.xlabel(r'$\beta$', fontsize=13)
plt.ylabel(r'$\lambda_{\min}$', fontsize=13)
plt.title("Convexity Window Comparison Across Lattice Sizes\nScale = 0.05",
plt.grid(True)
plt.legend(fontsize=12)
plt.tight_layout()
plt.show()
```

```
# =====
# FULL L = 8 CONVEXITY SURFACE SCAN
# Target: Map the Gribov Horizon at Thermodynamic Volume
# =====

import time
import jax.numpy as jnp

# Define the full grid
```

```
.. DEFINE THE FULL GRID
# We start at 0.4 (Strong Coupling) and go to 3.0 (Weak Coupling)
betas = jnp.linspace(0.4, 3.0, 8)
scales = (0.05, 0.10, 0.15)

print(f"\n### STARTING FULL L=8 SURFACE SCAN ###")
print(f"Grid: 8 Betas x 3 Scales x 3 Samples = 72 Eigensolves")
print(f"Est. Time on A100: ~6-10 minutes")

t0 = time.time()

# Execute using the loaded engine
# Note: c0=0.125 is the Correct SU(3) Haar Coefficient
res8_full = convexity_grid(8, betas, scales, c0=0.125, n_samples=3)

print("\n### L = 8 FULL SCAN COMPLETE ###")
print(f"Total Runtime: {time.time() - t0:.2f} seconds")

# --- DATA DUMP FOR PAPER ---
print("\n--- COPY THIS DATA FOR MANUSCRIPT ---")
print("beta    scale    lambda_min")
for b, s, lam in res8_full:
    print(f"{b:4.2f}    {s:5.3f}    {lam:+.6f}")
```

Next steps: [Explain error](#)

```
# =====
# FINAL MANUSCRIPT CLOSING: THEOREM 1(A) DYNAMIC RESTORATION
# Target: SU(3), L=8, beta=3.00, scale=0.15 (Deepest Instability)
# =====

import jax
import jax.numpy as jnp
import jax.lax as lax
import time

jax.config.update("jax_enable_x64", False)

# --- RE-USE FAST ENGINE COMPONENTS ---
def su3_generators():
    lam1 = jnp.array([[0,1,0],[1,0,0],[0,0,0]], jnp.complex64)
    lam2 = jnp.array([[0,-1j,0],[1j,0,0],[0,0,0]], jnp.complex64)
    lam3 = jnp.array([[1,0,0],[0,-1,0],[0,0,0]], jnp.complex64)
    lam4 = jnp.array([[0,0,1],[0,0,0],[1,0,0]], jnp.complex64)
    lam5 = jnp.array([[0,0,-1j],[0,0,0],[1j,0,0]], jnp.complex64)
    lam6 = jnp.array([[0,0,0],[0,0,1],[0,1,0]], jnp.complex64)
    lam7 = jnp.array([[0,0,0],[0,0,-1j],[0,1j,0]], jnp.complex64)
    lam8 = jnp.array([[1,0,0],[0,1,0],[0,0,-2]], jnp.complex64) / jnp.sqrt(3)
    return 1j * jnp.stack([lam1,lam2,lam3,lam4,lam5,lam6,lam7,lam8], 0) / 2.0

T_SU3 = su3_generators()

def su3_alg_from_vec(a):
    return jnp.einsum("...a,aij->...ij", a, T_SU3)

@jax.checkpoint
def su3_alg_from_vec_checkpoint(a):
```

```

def su3_exp_pade22(A):
    I = jnp.eye(3, dtype=jnp.complex64)
    A2 = A @ A
    c1, c2 = 0.5, 0.0833333336
    Num = I + c1*A + c2*A2
    Den = I - c1*A + c2*A2
    return jnp.linalg.solve(Den, Num)

def build_links_factory(L):
    @jax.checkpoint
    def build_links(params):
        flat = params.reshape(-1,8)
        A = jax.vmap(su3_alg_from_vec)(flat)
        return jax.vmap(su3_exp_pade22)(A).reshape(L,L,L,L,4,3,3)
    return build_links

# JIT-ISOLATED FOR SPEED
@jax.jit
def compute_plaquette_sum(U, beta):
    S = 0.0
    for mu in range(4):
        for nu in range(mu+1, 4):
            U_mu = U[..., mu, :, :]
            U_nu_shift = jnp.roll(U[..., nu, :, :], -1, axis=mu)
            U_mu_dag_shift = jnp.swapaxes(jnp.conjugate(jnp.roll(U[..., mu, :, :], -1, axis=mu)), -1, -2)
            U_nu_dag = jnp.swapaxes(jnp.conjugate(U[..., nu, :, :]), -1, -2)
            P = U_mu @ U_nu_shift @ U_mu_dag_shift @ U_nu_dag
            trP = jnp.real(jnp.einsum("...ii->...", P))
            S += jnp.sum(1.0 - trP/3.0)
    return beta * S

def vectorized_wilson_action(params, L, beta, build_links_L):
    U = build_links_L(params)
    return compute_plaquette_sum(U, beta)

@jax.jit
def haar_mass(params, c0):
    flat = params.reshape(-1,8)
    def per(a):
        A = su3_alg_from_vec(a)
        return jnp.real(jnp.trace(A.conj().T @ A))
    return c0 * jax.vmap(per)(flat).sum()

def make_flat_funcs(L, beta, c0):
    build_links_L = build_links_factory(L)
    def unflatten(x): return x.reshape((L,L,L,L,4,8))

    @jax.jit
    def flat_action(theta):
        params = unflatten(theta)
        return vectorized_wilson_action(params, L, beta, build_links_L) + haar

```

```

    return flat_action, unflatten

def hvp(flat_action, theta, v):
    g = jax.grad(flat_action)
    _, hv = jax.jvp(g, (theta,), (v,))
    return hv

def lanczos_min(flat_action, theta, k=25, seed=0):
    key = jax.random.PRNGKey(seed)
    n = theta.shape[0]
    v0 = jax.random.normal(key, (n,))
    v0 /= jnp.linalg.norm(v0)

    def step(carry, _):
        v_prev, v, beta_prev = carry
        w = hvp(flat_action, theta, v)
        w -= beta_prev * v_prev
        alpha = jnp.dot(w, v)
        w -= alpha * v
        beta = jnp.linalg.norm(w)
        v_next = w / (beta + 1e-9)
        return (v, v_next, beta), (alpha, beta)

    (_, _, _), (alphas, betas) = lax.scan(step, init=(jnp.zeros_like(v0), v0,
    alphas = jnp.array(alphas)
    betas = jnp.array(betas[:-1])
    T = jnp.diag(alphas) + jnp.diag(betas, 1) + jnp.diag(betas, -1)
    return float(jnp.linalg.eigvalsh(T)[0])

# =====
# RG FLOW INTEGRATOR
# =====
@jax.jit
def flow_step(theta, flat_action_grad, dt):
    grads = flat_action_grad(theta)
    return theta - dt * grads

def run_flow_test():
    # TARGET: The Deepest Instability from L=8 scan
    L = 8
    beta = 3.00
    scale = 0.15
    c0 = 0.125

    dt = 0.005
    steps = 30

    print(f"--- THEOREM 1(A) VERIFICATION: RG FLOW (L={L}) ---")
    print(f"Target: Restore convexity to unstable state (beta={beta}, scale={:

# 1 Initialize Engine

```

```

# 1. Initialize Engine
print("Compiling...")
flat_action, unflatten = make_flat_funcs(L, beta, c0)
flat_action_grad = jax.jit(jax.grad(flat_action))

# 2. Create the Unstable State
key = jax.random.PRNGKey(101)
theta = (scale * jax.random.normal(key, (L,L,L,L,4,8))).reshape(-1)

# 3. Measure Initial State
t0 = time.time()
lam = lanczos_min(flat_action, theta, k=25)
print(f"Step 0 (t=0.000): lambda_min = {lam:+.6f} [UNSTABLE]")

# 4. Apply Flow
curr_theta = theta
for i in range(1, steps+1):
    curr_theta = flow_step(curr_theta, flat_action_grad, dt)

    # Measure every 2 steps
    if i % 2 == 0:
        lam = lanczos_min(flat_action, curr_theta, k=25)
        status = " [RESTORED]" if lam > 0 else " [UNSTABLE]"
        print(f"Step {i:2d} (t={i*dt:.3f}): lambda_min = {lam:+.6f}{status}")

        if lam > 0:
            print(f"\n>>> SUCCESS: Mass gap restored at Flow Time t = {i*dt:.3f}s")
            break

print(f"Total Time: {time.time()-t0:.2f}s")

if __name__ == "__main__":
    run_flow_test()

```

Next steps:

[Explain error](#)